Application of Speaker Diarization on Real-world, Noisy Audio

Sebastian Joseph
The University of Texas at Austin
sebaj@utexas.edu

Younghan Park
The University of Texas at Austin
younghanpark@utexas.edu

Long Do
The University of Texas at Austin
lhd379@utexas.edu

Abstract

Speaker diarization, the process of segmenting and co-indexing audio recordings by speakers, has become increasingly crucial in handling the vast volume of audio data generated from broadcasts, meetings, and voice mails. However, the conventional application of speaker diarization faces challenges, particularly in real-world scenarios impair the quality of results. In this project, we address the limitations of speaker diarization on noisy data through a comprehensive approach. Our study is structured into three main objectives. Firstly, we categorize various types of noise, including background noise, overlapping speakers, and distortions, and develop methods to synthetically recreate these in clean audio. Secondly, we enhance the robustness of speaker diarization to noise by implementing techniques such as audio source separation, speech enhancement algorithms, and other noise reduction strategies. Finally, we explore novel ways to evaluate speaker diarization performance on noisy data, introducing specific metrics and conducting experiments on noise-augmented datasets. This project aims to significantly contribute to the advancement of speaker diarization methods, particularly in the context of handling noisy audio data, with potential applications in accurate transcription of multi-speaker audio in real-world settings.

1 Introduction

In recent years, the proliferation of audio data from diverse sources, including broadcasts [1], meetings [2], and voice mails, has underscored the importance of effective speaker diarization. This task, involving the segmentation and indexing of audio recordings by speakers, is pivotal for applications such as transcription, information retrieval, and content summarization. However, the practical application of speaker diarization encounters formidable challenges [3], particularly when confronted with real-world scenarios characterized by background noise, overlapping speakers, and other distortions that compromise the fidelity of speaker audio.

Recognizing the limitations of conventional speaker diarization approaches, our project endeavors to enhance the robustness of these methods in the presence of noise. The first phase of our study involves a systematic categorization of different types of noise, ranging from background noise to overlapping speakers and distortions. Building upon this categorization, we develop methods to synthetically recreate these noise scenarios in clean audio, providing a controlled environment for subsequent experimentation.

Subsequently, we address the core issue of noise interference by implementing a series of techniques aimed at fortifying speaker diarization against such challenges. These techniques include the integration of an audio source separation pipeline into the diarization process, the application of diverse speech enhancement algorithms, and the exploration of other noise reduction strategies. Through

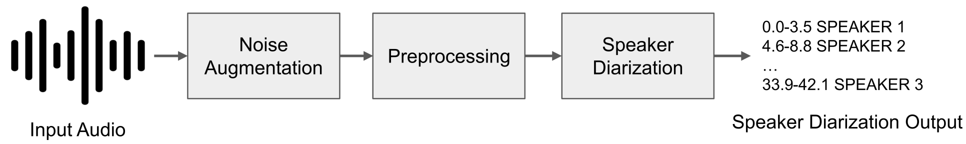

Figure 1: Visualization depicting the entire framework process. Noise augmentation and preprocessing steps are optional components. Following the creation of the input audio with these optional steps, the diarization pipeline is executed, and subsequent evaluation is conducted.

these interventions, we aim to significantly improve the accuracy and reliability of speaker diarization outcomes on noisy data.

The concluding phase of our project is dedicated to determining the optimal combination of preprocessing methods for handling noisy audio. Our approach involves conducting thorough experiments using datasets augmented with noise. To achieve this, we will employ a state-of-the-art model in the field of speaker diarization and carry out comprehensive experiments to assess the effectiveness of various preprocessing techniques.

By addressing the challenges posed by noise in speaker diarization, our project aims to contribute substantially to the refinement of existing methods, opening avenues for more accurate transcription of multi-speaker audio in real-world applications.

2 Background

Figure 1 clearly describes our framework.

2.1 Speaker Diarization

Speaker diarization is the process of partitioning an audio stream containing human speech into homogeneous segments according to the identity of each speaker [4]. Simply put, this process addresses the question of "who spoke when" [5] by logging speaker-specific salient events on multi-speaker audio data.

The performance of the speaker diarization system can be measured in various ways, such as diarization error rate (DER) [6] or Jaccard error rate, which was first introduced in [7]. In this work, we will employ DER as our evaluation metric:

$$\text{DER} = \frac{\text{false alarm} + \text{missed detection} + \text{confusion}}{\text{total}}$$

Here, "false alarm" represents the duration of non-speech incorrectly classified as speech, "missed detection" is the duration of speech incorrectly classified as non-speech, "confusion" is the duration of speaker confusion, and "total" is the sum over all speakers of their reference speech duration.

2.2 Noise in Audio

Much of the signals used to train and build audio diarization systems are of high studio quality, simply because it is easier to gather these signals and get the corresponding ground truth, human-annotated labels with high reliability. However, this reliance on high-quality data does make it difficult for such systems to generalize to noisier signals that are more prevalent in the real world.

The DIHARD Challenge [8, 7] was created to address this problem. It is a benchmark to designed to test the robustness of diarization systems. The dataset associated with the challenge introduces variations in audio equipment, noise conditions, and conversational domain. Despite these efforts to improve the robustness of speaker diarization systems, the amount of noisy, gold-labeled data available is quite small compared to wide range of studio quality data that is available for the task of speaker diarization.

Our work focuses on creating noisy, labeled data by synthetically adding noise to clean, studio-quality data. This methodology allows for a simulated version of real-world, noisy audio to be created a large scale on existing data. In this paper, we introduce two methods to noise clean data: overlaying background noise and simulating crosstalk.

2.3 Audio Source Separation

Audio source separation is the separation of a set of source signals from a set of mixed signals, without or with limited information about the source signals or the mixing process. This is also known as the "cocktail party problem". Formally, in this problem, the waveform of the observed signal $x(t)$ and the independence between the signal sources are used to make the estimated signal $s^*(t) = \{s_1^*(t), s_2^*(t), ..., s_n^*(t)\}$ as close to the signal source $s(t) = \{s_1(t), s_2(t), ..., s_n(t)\}$ as possible. The observed mixed signal is $x(t) = \{x_1(t), x_2(t), ..., x_m(t)\}$. The mathematical model of audio source separation is a linear instantaneous mixture model: $x(t) = As(t) + n(t)$, where A is the mixing matrix, m represents the number of source signals, and n represents the number of receiving antenna elements. When $n < m$, it is defined as under-determined source separation. When $n = 1$, it is defined as single-channel source separation under under-determined conditions. In this project, we experiment with three audio source separation methods: Independent Component Analysis (ICA), Non-Negative Matrix Factorization (NMF), and Wave-U-Net.

2.3.1 Independent Component Analysis (ICA)

Independent Component Analysis (ICA) is a widely used blind source separation technique in the field of data analysis that allows researchers to separate and identify the underlying independent sources in a multivariate data set. The general framework for this can be represented as $x = As$, where A is an unknown square mixing matrix [9]. We have the observation x and our goal is to recover the source s. Let $W = A^{-1}$ be the unmixing matrix. Our goal is to find W, so given the observation x, we can recover the sources s by computing $s = Wx$. There are several algorithms for decreasing the mutual information between signals; however, for this project, we use the gradient descent method [10].

Algorithm 1 ICA Gradient Descent

```
Assume X = AU
Initialize the un-mixing matrix W with small random values
i ← 0
while i < max_iter do
    Calculate Y = WX
    Calculate Z = 1/(1 + e^{-Y})
    find ΔW = η(I + (1 − 2Z)Y^T)W where η is a small learning rate
    Update W = W + ΔW
end while
Return W
```

2.3.2 Non-Negative Matrix Factorization (NMF)

Non-Negative Matrix Factorization (NMF) is a matrix factorization technique that decomposes a non-negative matrix A into a pair of non-negative matrices B and C with lower rank such that $A \approx BC$. Source separation can be viewed as a matrix factorization problem, where the source mixture is modeled as a matrix containing its spectrogram representation [11]. The spectrogram of a signal $s(t)$ can be estimated by computing the squared magnitude of the short-time Fourier transform (STFT) of the signal $s(t)$; thus, $s(t)$ can be recovered from the spectrogram through the inverse short-time Fourier transform (ISTFT) after processing the signal spectral domain [12].

2.3.3 Wave-U-Net

The last method we consider is a deep learning separation method. The Wave-U-Net is an end-to-end convolutional neural network applicable to audio source separation tasks without pre-processing or post-processing. The model processes the time series context by repeatedly performing down-sampling and convolution of feature maps to combine high and low-level features at different time-scales [14]. However, one of the drawbacks of Wave-U-Net is that in each feature map generated by convolution, the sampling rate of the original signal is used as the resolution; thus, the memory

Algorithm 2 NMF with β-Divergence and Multiplicative Updates [13]

Initialize H, W with nonnegative random coefficients
$i \leftarrow 0$
while $i < max\_iter$ **do**
  Update $H = H \odot \frac{W^T[(WH)^{\beta-2} \odot V]}{W^T(WH)^{\beta-1}}$
  Update $W = W \odot \frac{[(WH)^{\beta-2} \odot V]H^T}{(WH)^{\beta-1}H^T}$
end while
return H, W

consumption required is high [15]. In this project, instead of using the top-of-the-shelf Wave-U-Net architecture and training it from scratch, we use the pre-trained model.

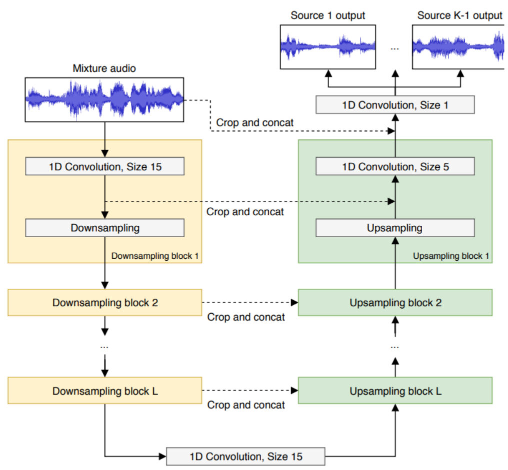

Figure 2: Wave-U-Net with K sources and L layers

2.4 Speech Enhancement

Speech enhancement is a signal-processing task that involves improving the quality of speech signals captured under noisy or degraded conditions and making them clearer. Speech enhancement is a natural application of source separation.

Suppose $y(t)$, $s(t)$, and $d(t)$ are the sampled noisy speech, clean speech, and additive noise, respectively, from this, their relationship is $y(t) = s(t) + d(t)(1)$ and our goal is to extract $s(t)$ given the observation $y(t)$.

In this project, we employ two of the well-established methods in the speech enhancement literature, Spectral Subtraction and Wiener Filter, in our pipeline and conduct experiments with them.

2.4.1 Spectral Subtraction

The spectral subtraction is one of the first algorithms proposed for speech enhancement tasks. For this method, to estimate clean speech, we estimate the noise spectrum and subtract it from the noisy speech spectrum. However, the drawback of spectral subtraction is the presence of processing distortions.

Performance Fourier transforms on $y(y)$, $s(t)$, and $d(t)$, then the representation of (1) in the short-time Fourier transform (STFT) is given by $Y(\omega) = S(\omega) + D(\omega)$. Assuming that pure speech signal and noise signal are not correlated, we get $|Y(\omega)|^2 = |S(\omega)|^2 + |D(\omega)|^2$. Hence, speech can be estimated by subtracting a noise estimate from the received signal $|\hat{S}(\omega)|^2 = |Y(\omega)|^2 - |\hat{D}(\omega)|^2$ and the estimation of the noise spectrum can be obtained by averaging recent speech pauses frames $|\hat{D}(\omega)|^2 = \frac{1}{M}\sum_{j=0}^{M-1}|Y_{SP_j}(\omega)|^2$ [16].

2.4.2 Wiener Filtering

The Wiener Filter is an optimal filter that minimizes the mean-squared error between the original and enhanced speech. The method is implemented in the frequency domain and depends on the filter transfer function from sample to sample based on the speech signal statistics; the local mean and variance. Similar to Spectral Subtraction, it assumes speech and noise are not correlated. The gain function of the Wiener filter can be expressed as $H_{wiener}(\omega) = \frac{P_s(\omega)}{P_s(\omega)+P_d(\omega)}$ where $P_s(\omega)$ and $P_d(\omega)$ are the power spectral density of clean speech and noise, respectively. Thus clean speech can be estimated as $|\hat{S}(\omega)|^2 = H_{wiener}(\omega)|Y(\omega)|^2$ [16].

3 Experiments

3.1 Experimental Setup

For tasks requiring GPU, a single NVIDIA A100 GPU is employed.

Dataset We use the ICSI dataset [2] as the main dataset for our experiments. The dataset contains around 70 hours of speech collected from meeting recordings at various business and academic institutions. These recordings are collected from meeting rooms with a multichannel, studio-quality recording systems. For each recording, there are associated segment annotations, describing the start and end time of when a speaker is talking within the recording. These segment annotations are initially separated according to speaker. However, we processed these annotations into a single file, ordering segments according to their start time.

Noise Augmentation We augment the clean, studio quality data from the ICSI data with two different types of noise: background noise and crosstalk.

- **Background Noise:** This method of augmentation involves overlaying music or noise over signal data. We use three minute background noise samples from copyright-free music and overlay them over the meeting signal in a loop. In order to avoid drowning out the speech in the signal, the background noise signal is normalized as to not exceed the loudness of the speech signal.
- **Crosstalk:** In order to simulate crosstalk, we manipulate both the ground truth segment annotations and the corresponding speech signal. First, we analyze the segment annotations and collect pairs of segments that are temporally close but yet do not overlap. These segments are then manipulated so that they overlap by a specified overlap window. After a pair of segments are initially manipulated, it is important to propagate the change in time to segments ahead in time as to ensure their correctness. Finally, based on this modified ground truth, we manipulate the speech signals to represent these added overlaps by splitting the audio at the time of overlap and overlaying the split signal. The end result is a speech signal with simulated crosstalk and segment annotations that accurately correspond to this modified signal.

We used the pydub Python library to manipulate the audio signals [1]. We also manually verify that the speech signals are still audible and speakers can still distinguished by humans after our augmentations.

Preprocessing In the pre-processing step, we feed the noisy data to the Audio Source Separation and Speech Enhancement pipeline. We first use Audio Source Separation methods to separate speech from noise. However, the produced speech from Audio Source Separation still contains some noise; thus, we apply Speech Enhancement techniques to reduce noise. As mentioned in section 2.3, we will use 3 methods to separate clean speech from noise and employ 2 methods to enhance clean speech as mentioned in section 2.4. Additionally, we also try to experiment with using only Audio Source Separation and using only Speech Enhancement only. Thus, there are 11 experimental setups in total.

Despite having 11 experimental setups, we only reported 1 experimental setup where we use Wave-U-Net for audio source separation and Spectral Subtraction for Speech Enhancement. When running

[1] Details about the pydub library can be found here: https://github.com/jiaaro/pydub

Diarization Error Rate (DER)		Preprocessing Methods			
		None	Audio Source Separation	Speech Enhancement	Both
Added Noise	None	**0.390**	0.591	0.449	0.637
	Crosstalk	**0.404**	0.610	0.465	0.651
	Background Music	**0.744**	1.653	1.139	1.851

Table 1: Diarization Error Rate (DER) across various settings, with each row representing a distinct augmented noise type and each column corresponding to different combinations of preprocessing methods. Bold numbers highlight the optimal preprocessing methods for specific noise types.

Name	Situation	Clean
AISHELL [19]	talk	✓
AliMeeting [20]	meeting	
AMI [21]	meeting	
AVA-AVD [22]	various	
DIHARD [7]	various	
Ego4D [23]	various	
MSDWild [24]	vlog	
REPERE [25]	TV show	
VoxConverse [26]	debate	

Table 2: Distribution of the datasets that were used in training the checkpoint that we use

the ICA, NMF, and Wave-U-Net on a subset of noisy ICSI dataset that we created, we noticed that Wave-U-Net outperforms ICA and NMF by a huge margin. Output produced NMF contains a lot of static noise and output produced by ICA cannot cleanly separate speech from music. For Speech Enhancement, we decided to use Spectral Subtraction because it produce a robust speech-to-noise ratio in adversely conditions so that Pyannote can be used in real-life situations. In case of Wiener Filter because the method minimizes the MSE under a series of assumptions, which usually do not fit in real-life scenario. In other word, in practice, it is nearly impossible to yield the solution with the smallest mean squared error (MSE) possible.

Evaluation For speaker diarization, we use pyannote.audio 3.0 [17] [18], which is one of the current state-of-the-art speaker diarization model. The checkpoint that we use is trained with a combination of various datasets, AISHELL [19], AliMeeting [20], AMI [21], AVA-AVD [22], DIHARD [7], Ego4D [23], MSDWild [24], REPERE [25], and VoxConverse [26]. The evaluation of diarization performance is conducted using the `Annotation` data structure from [27], which facilitates the computation of Diarization Error Rate (DER).

Before initiating this process, a preliminary step involves the implementation of a function to consolidate multiple ground truth annotations, particularly for the ICSI dataset, where annotations are distributed across multiple files, each representing speaking segments of a specific individual in a meeting. These consolidated annotations are then formatted into a single `rttm` file, aligning with the output format of the diarization pipeline.

3.2 Results

Table 1 presents our primary findings. Notably, among various types of augmented noise, synthetically overlapping utterances exhibit minimal impact on Diarization Error Rate (DER). Conversely, introducing background music into the original audio yields a substantial reduction in DER. In relation to preprocessing methods, our experiments indicate an overall deterioration in DER across all methods tested. Notably, audio source separation exhibits a significant worsening of DER compared to speech enhancement. Combining both audio source separation and speech enhancement results in the highest DER observed in our study.

4 Discussion

In this section, we will delve into an analysis of the factors contributing to our project's challenges and outcomes. Table 2 illustrates the distribution of the data used to train the employed checkpoint. The table indicates a diverse range of situations for speech, with many distributions containing various types of noise. Consequently, we anticipated that the combination of the model and checkpoint we used already possessed the capability not only to segment speech but also to detect speech. However, the application of a multi-stage pipeline involving noise augmentation and preprocessing might resulted in unexpected side effects that harmed audio quality. This unforeseen impact stemmed from encountering unseen types of noise during the model training process.

Our results, unfortunately, are not as we expected. The DER of our pipeline is worse than the baseline Pyannote model. However, we believe there are several reasons to explain this behavior. Firstly, we suspect that Pyannote is trained with noisy data; thus, it is robust to noise. Therefore, the baseline Pyannote model still achieves good DER even with noisy data. The Whisper model shows that the model can be robust to noise if trained with noisy and diverse data [28]. Secondly, there are several weaknesses for each audio separation and speech enhancement method that we use. ICA cannot be applied to just any mixed signal. The mixed signals must be a linear combination of source signals. Most importantly, the source signals have to be non-Gaussian. The disadvantage of NMF is it does not produce a unique solution, therefore introducing the difficulty in choosing the representative reconstructions from several possible outcomes. Additionally, since only nonnegative mixtures and nonnegative mixing operations are allowed, NMF is a poor method for recovering mixed-sign signals and estimating mixed-sign mixing matrices [29]. Lastly, for Wave-U-Net, this model is proposed to separate singing voice and accompaniment. In our case, we use the ICSI dataset and add random music to it; thus, the clean speech is not the singing voice. Therefore, this might be one of the reasons why the performance of our pipeline is worse than the baseline Pyannote.

5 Conclusion

In conclusion, our project aimed to address the challenge of noise in speaker diarization, recognizing its significance in managing diverse audio data. Despite our systematic efforts, including the implementation of targeted techniques such as audio source separation and speech enhancement algorithms, we acknowledge that we faced challenges in significantly improving the performance in real-world scenarios.

While our efforts did not lead to the expected advancements, this project provides valuable insights into the limitations and complexities inherent in addressing noise in audio data. Acknowledging these challenges is crucial for future endeavors in refining speaker diarization techniques. Our findings contribute to the ongoing dialogue within the field, emphasizing the need for continued exploration and innovation to overcome the persistent hurdles posed by noise in practical applications.

Building upon the insights gained from this project, future work can explore several avenues to further advance the robustness of speaker diarization in the presence of noise. Some potential directions include:

- Advanced Preprocessing Techniques: Future work could delve into the exploration and development of sophisticated preprocessing methods tailored to address the challenges presented by diverse noise types. This exploration may include the integration of machine learning approaches or novel signal processing techniques. Additionally, concatenating preexisting neural models, such as prepending a neural audio source separation module to a speaker diarization model, offers a promising avenue for enhancing the overall process. By adopting this approach, several strategies can be employed, including training both modules jointly from scratch, retaining parameters of the diarization pipeline while fine-tuning the audio source separation module, and more. These investigations have the potential to further elevate the effectiveness and adaptability of the diarization pipeline in handling complex acoustic environments.
- Adaptive Models: An intriguing avenue for future work involves exploring the feasibility of adaptive speaker diarization models capable of dynamically adjusting their parameters in response to the characteristics of input audio, particularly in the presence of varying levels and types of noise. One potential strategy is to condition the model on a general-purpose

audio embedding during training. In the inference phase, the framework can extract the general-purpose embedding from the input audio, subsequently utilizing different sets of parameters to compute the diarization output. This adaptive approach holds promise for enhancing the model's adaptability to diverse acoustic environments, ultimately contributing to more robust and accurate speaker diarization outcomes.

- Real-world Datasets: While this work successfully synthesized various types of noise that closely resemble real-world conditions, it's important to acknowledge the inherent distinction between synthetic and authentic noise. Moving forward, a valuable avenue for research involves the exploration and utilization of larger, more diverse real-world datasets. This initiative aims to capture a broader spectrum of noise scenarios, providing a more comprehensive understanding of the complexities encountered in practical applications. Acquiring and analyzing such datasets can contribute to refining the robustness of models and methodologies, ensuring their effectiveness in real-world, dynamic environments.
- Integration with Multimodal Data: In real-world situations where discerning speakers becomes challenging due to excessive noise, individuals often rely on information from alternate modalities, such as vision. Recognizing this, there is potential to delve deeper into the integration of additional modalities, including visual or contextual information, to enhance audio-based speaker diarization. This exploration aims to improve performance in challenging scenarios, where traditional audio cues may be insufficient. Leveraging complementary information from multiple modalities holds promise for achieving more robust and accurate speaker diarization outcomes in diverse and dynamic environments.

By exploring these future avenues, people in the field of speaker diarization can continue to refine speaker diarization techniques, ultimately contributing to more effective and reliable applications in transcription, information retrieval, and content summarization in diverse and noisy audio environments.

References

[1] Anthony Larcher, Ambuj Mehrish, Marie Tahon, Sylvain Meignier, Jean Carrive, David Doukhan, Olivier Galibert, and Nicholas Evans. Speaker embeddings for diarization of broadcast data in the allies challenge. In *ICASSP*, 2021.

[2] A. Janin, D. Baron, J. Edwards, D. Ellis, D. Gelbart, Nathaniel Morgan, B. Peskin, Thilo Pfau, Elizabeth Shriberg, A. Stolcke, and Chuck Wooters. The icsi meeting corpus. In *ICASSP*, 2003.

[3] Joon Son Chung, Bong-Jin Lee, and Icksang Han. Who Said That?: Audio-Visual Speaker Diarisation of Real-World Meetings. In *INTERSPEECH*, 2019.

[4] Md Sahidullah, Jose Patino, Samuele Cornell, Ruiqing Yin, Sunit Sivasankaran, Hervé Bredin, Pavel Korshunov, Alessio Brutti, Romain Serizel, Emmanuel Vincent, Nicholas Evans, Sébastien Marcel, Stefano Squartini, and Claude Barras. The speed submission to dihard ii: Contributions & lessons learned. *arXiv preprint arXiv:1911.02388*, 2019.

[5] Tae Jin Park, Naoyuki Kanda, Dimitrios Dimitriadis, Kyu J. Han, Shinji Watanabe, and Shrikanth Narayanan. A review of speaker diarization: Recent advances with deep learning. *Computer Speech & Language*, 2022.

[6] Jonathan G. Fiscus, Jerome Ajot, Martial Michel, and John S. Garofolo. The rich transcription 2006 spring meeting recognition evaluation. In *Machine Learning for Multimodal Interaction*, 2006.

[7] Neville Ryant, Kenneth Church, Christopher Cieri, Alejandrina Cristia, Jun Du, Sriram Ganapathy, and Mark Liberman. The second dihard diarization challenge: Dataset, task, and baselines. *arXiv preprint arXiv:1906.07839*, 2019.

[8] Neville Ryant, Kenneth Church, Christopher Cieri, Alejandrina Cristia, Jun Du, Sriram Ganapathy, and Mark Liberman. First dihard challenge evaluation plan. In *INTERSPEECH*, 2018.

[9] Jens Wellhausen and Volker Gnann. Independent component analysis for audio signal separation. In *Multimedia Systems and Applications*, 2016.

[10] Andrew Ng. *CS229 Lecture notes Part XII Independent Components Analysis*, 2020. URL `https://cs229.stanford.edu/notes2020spring/cs229-notes11.pdf`.

[11] Wisam Reid, Kai-Chieh Huang, and Doron Roberts-Kedes. Blind audio source separation pipeline and algorithm evaluation. In *Stanford CS229*, 2015.

[12] Joan Bruna, Pablo Sprechmann, and Yann LeCun. Source separation with scattering non-negative matrix factorization. In *ICASSP*, 2015.

[13] Romain Serizel, Slim Essid, and Gaël Richard. Mini-batch stochastic approaches for accelerated multiplicative updates in nonnegative matrix factorisation with beta-divergence. In *MLSP*, 2016.

[14] Simon Dixon Daniel Stoller, Sebastian Ewert. Wave-u-net: A multi-scale neural network for end-to-end audio source separation. *arXiv preprint arXiv:1806.03185*, 2018.

[15] Zhao M, Yao X, Wang J, Yan Y, Gao X, and Fan Y. Single-channel blind source separation of spatial aliasing signal based on stacked-lstm. *Sensors*, 2021.

[16] Navneet Upadhyay and Abhijit Karmakar. Speech enhancement using spectral subtraction-type algorithms: A comparison and simulation study. *Procedia Computer Science*, 2015.

[17] Alexis Plaquet and Hervé Bredin. Powerset multi-class cross entropy loss for neural speaker diarization. In *INTERSPEECH*, 2023.

[18] Hervé Bredin. pyannote.audio 2.1 speaker diarization pipeline: principle, benchmark, and recipe. In *INTERSPEECH*, 2023.

[19] Hui Bu, Jiayu Du, Xingyu Na, Bengu Wu, and Hao Zheng. Aishell-1: An open-source mandarin speech corpus and a speech recognition baseline. In *Oriental COCOSDA*, 2017.

[20] Fan Yu, Shiliang Zhang, Yihui Fu, Lei Xie, Siqi Zheng, Zhihao Du, Weilong Huang, Pengcheng Guo, Zhijie Yan, Bin Ma, Xin Xu, and Hui Bu. M2MeT: The ICASSP 2022 multi-channel multi-party meeting transcription challenge. In *ICASSP*, 2022.

[21] Jean Carletta, Simone Ashby, Sebastien Bourban, Mike Flynn, Mael Guillemot, Thomas Hain, Jaroslav Kadlec, Vasilis Karaiskos, Wessel Kraaij, Melissa Kronenthal, Guillaume Lathoud, Mike Lincoln, Agnes Lisowska, Iain McCowan, Wilfried Post, Dennis Reidsma, and Pierre Wellner. The ami meeting corpus: A pre-announcement. In *Machine Learning for Multimodal Interaction*, 2006.

[22] Eric Zhongcong Xu, Zeyang Song, Satoshi Tsutsui, Chao Feng, Mang Ye, and Mike Zheng Shou. Ava-avd: Audio-visual speaker diarization in the wild. In *ACM MM*, 2022.

[23] Kristen Grauman, Andrew Westbury, Eugene Byrne, Zachary Chavis, Antonino Furnari, Rohit Girdhar, Jackson Hamburger, Hao Jiang, Miao Liu, Xingyu Liu, Miguel Martin, Tushar Nagarajan, Ilija Radosavovic, Santhosh Kumar Ramakrishnan, Fiona Ryan, Jayant Sharma, Michael Wray, Mengmeng Xu, Eric Zhongcong Xu, Chen Zhao, Siddhant Bansal, Dhruv Batra, Vincent Cartillier, Sean Crane, Tien Do, Morrie Doulaty, Akshay Erapalli, Christoph Feichtenhofer, Adriano Fragomeni, Qichen Fu, Christian Fuegen, Abrham Gebreselasie, Cristina Gonzalez, James Hillis, Xuhua Huang, Yifei Huang, Wenqi Jia, Weslie Khoo, Jachym Kolar, Satwik Kottur, Anurag Kumar, Federico Landini, Chao Li, Yanghao Li, Zhenqiang Li, Karttikeya Mangalam, Raghava Modhugu, Jonathan Munro, Tullie Murrell, Takumi Nishiyasu, Will Price, Paola Ruiz Puentes, Merey Ramazanova, Leda Sari, Kiran Somasundaram, Audrey Southerland, Yusuke Sugano, Ruijie Tao, Minh Vo, Yuchen Wang, Xindi Wu, Takuma Yagi, Yunyi Zhu, Pablo Arbelaez, David Crandall, Dima Damen, Giovanni Maria Farinella, Bernard Ghanem, Vamsi Krishna Ithapu, C. V. Jawahar, Hanbyul Joo, Kris Kitani, Haizhou Li, Richard Newcombe, Aude Oliva, Hyun Soo Park, James M. Rehg, Yoichi Sato, Jianbo Shi, Mike Zheng Shou, Antonio Torralba, Lorenzo Torresani, Mingfei Yan, and Jitendra Malik. Ego4d: Around the World in 3,000 Hours of Egocentric Video. In *CVPR*, 2022.

[24] Tao Liu, Shuai Fan, Xu Xiang, Hongbo Song, Shaoxiong Lin, Jiaqi Sun, Tianyuan Han, Siyuan Chen, Binwei Yao, Sen Liu, Yifei Wu, Yanmin Qian, and Kai Yu. MSDWild: Multi-modal Speaker Diarization Dataset in the Wild. In *INTERSPEECH*, 2022.

[25] Aude Giraudel, Matthieu Carré, Valérie Mapelli, Juliette Kahn, Olivier Galibert, and Ludovic Quintard. The REPERE corpus : a multimodal corpus for person recognition. In *LREC*, 2012.

[26] Joon Son Chung, Jaesung Huh, Arsha Nagrani, Triantafyllos Afouras, and Andrew Zisserman. Spot the conversation: speaker diarisation in the wild. In *INTERSPEECH*, 2020.

[27] Hervé Bredin. pyannote.metrics: a toolkit for reproducible evaluation, diagnostic, and error analysis of speaker diarization systems. In *INTERSPEECH*, 2017.

[28] Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. Robust speech recognition via large-scale weak supervision. In *ICML*, 2023.

[29] Andr iMirzal. Nmf versus ica for blind source separation. In *Advances in Data Analysis and Classification*, 2014.